CUAD Final Data Exploration & Chunking

Implementation guide and tokenization handoff | October 2026

1. Purpose and source notebooks

This guide accompanies the clean notebook. It preserves the original CUAD exploration from the uploaded notebook and replaces experimental Version 1 / Version 2 chunking with the Version 3 algorithm from the uploaded comparison notebook. The original notebooks remain unchanged.

2. Data exploration retained

The original analysis reports 510 contracts, 41 CUAD clause categories, and 20,910 contract-category checks. Of these, 6,702 were answerable and 14,208 unanswerable. Class frequencies vary substantially. The supplied split contains 408 training contracts and 102 test contracts; the notebook checks contract-title overlap.

3. Why fixed windows were insufficient

The first method used 300-word chunks with 50-word overlap. Some annotated spans crossed boundaries. Additional answer windows improved coverage, but many chunks still partially overlapped annotations. A partial overlap can create a misleading negative or an ambiguous positive. Version 2 used more overlap but retained many partial cases.

4. Final Version 3 logic

• Reconstruct each training contract by joining paragraph contexts with exactly one space.

• Convert answer_start offsets from paragraph-relative positions into contract-relative character offsets using the identical separator. Verify every extracted span matches its original answer text.

• Find word boundaries with regular expressions. Start with a 300-word window and 50-word overlap.

• If the end boundary falls inside an annotated answer, extend it to that answer end; repeat because extension can encounter another annotation. Move to complete word boundaries.

• If the start boundary cuts an annotated answer, move it backward to include the complete answer.

• Assign every fully contained normalized clause category; multi-label chunks are expected.

• Audit each chunk for exact text, expected labels and any partial answer. Exclude chunks with any partial overlap, then independently check coverage of every annotated answer.

5. Important correction found during consolidation

The uploaded Version 3 function joined paragraphs with newline characters, whereas its annotation offsets were computed with spaces. This inconsistency could misalign offsets. The final notebook consistently uses spaces for both and asserts that all source annotations match. Consequently, previous numeric results should be treated as historical until the final notebook is executed on the CUAD data files.

6. Historical results (before the correction)

Metric	Observed result
V1 strict answer coverage	69.02%
V2 strict answer coverage	75.59%
V3 strict answer coverage	100% (11,180 / 11,180)
V3 candidate chunks	13,005
V3 partial-overlap chunks	2
V3 longest chunk	1,098 words
V3 average chunk	308 words

These are prior notebook outputs, not newly validated final results. The corrected notebook recomputes all metrics and stops if source positions, text, labels or full-answer coverage fail.

7. Remaining limitations

Some chunks are longer than 300 words because answer boundaries take priority. Check token counts using the chosen model tokenizer and avoid truncating labeled evidence. Identical chunk text can appear multiple times, potentially with different labels; inspect conflicts before deduplication. Annotation-aware chunking uses ground-truth answers and is therefore suitable for training preparation only; test/inference chunking must not use annotations. The 100% coverage measure means every annotated answer is contained somewhere, not that all chunks are ideal.

8. Tokenization handoff

Run the final notebook top-to-bottom. It saves ../data/processed/final_training_chunks.json and cuad_categories.json. In the next notebook, select a tokenizer and model, inspect token length distribution, prepare multi-hot labels, and choose a model-compatible maximum length. Reserve validation contracts before model fitting and do not split overlapping chunks randomly across train and validation. Keep test contracts untouched until final evaluation.

9. GitHub recommendation

Commit the clean final notebook and this guide. Keep exploratory notebooks outside the tracked project or exclude them with precise .gitignore paths; verify git status before pushing. Do not accidentally ignore the final notebook. Dataset files may be too large or restricted for repository distribution; follow project data-handling rules.

Note: This PDF describes the complete methodology and decisions; the executable code is in the accompanying .ipynb file.